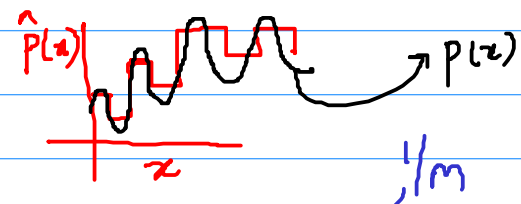


Histogram as a PDF estimator

Consider $X_1, X_2, \dots, X_n$ from unknown PDF $p(x)$, with domain $[0, 1]$. Assume a smooth PDF, i.e. $|p'(x)| \leq L < \infty$.

The hist. partitions $[0, 1]$ into M equal-width bins. $B_1 = [0, \frac{1}{m})$, $B_2 = [\frac{1}{m}, \frac{2}{m}) \dots B_{m-1} = [\frac{m-2}{m}, \frac{m-1}{m})$
 $B_m = [\frac{m-1}{m}, 1]$

For any value $x \in B_\ell$, 

$$\hat{p}_n(x) = \frac{\# \text{ of observations in } B_\ell}{n} \times \frac{1}{\text{binwidth}}$$

$$= \frac{M}{n} \sum_{i=1}^n \mathbb{I}(X_i \in B_\ell)$$

We are assuming a constant density inside any bin.

$$E(\hat{p}_n(x)) = \frac{M}{n} \sum_{i=1}^n E[\mathbb{I}(X_i \in B_\ell)] = P(X_i \in B_\ell)$$

$$= \frac{M}{n} \times \int_{(l-1)/m}^{l/m} p(u) du$$

$$= M \left[F_x(l/m) - F_x\left(\frac{l-1}{m}\right) \right]$$

$$= \frac{F_x(l/m) - F_x\left(\frac{l-1}{m}\right)}{\frac{l}{m} - \frac{l-1}{m}}$$

$$= p(x^*) \text{ for some } x^* \in \left[\frac{l-1}{m}, \frac{l}{m}\right)$$

Using Mean Value theorem

$$F_x\left(\frac{l}{m}\right) = F_x\left(\frac{l-1}{m}\right) + \frac{1}{m} p(x^*)$$

$$\boxed{\begin{aligned} g(x) &= g(x_0) \\ &+ (x - x_0) g'(x^*) \\ x^* &\in [x_0, x] \end{aligned}}$$

$$\begin{aligned}
 \text{bias}(\hat{p}_n(x)) &= E(\hat{p}_n(x)) - p(x) \\
 &= p(x^*) - p(x) \\
 &= p'(x^{**})(x^* - x) \\
 &\leq L(x^* - x) \leq L/M
 \end{aligned}$$

As $L \uparrow$, bias $\uparrow \equiv$ less smooth density
 As $M \uparrow$, bias $\downarrow$

$$\text{Var}(\hat{p}_n(x)) = \frac{M^2}{n^2} \sum_{i=1}^n \text{Var}(\underbrace{I(X_i \in B_e)})$$

This is a Bernoulli R.V. with success prob. $P(X_i \in B_e)$

$$\begin{aligned}
 &= \frac{M^2}{n^2} \times n \times \underbrace{P(X_i \in B_e)(1 - P(X_i \in B_e))} \\
 &= \frac{\frac{n^2}{M^2}}{n} \frac{p(x^*)}{\cancel{M}} \left(1 - \frac{p(x^*)}{M}\right) \\
 &\leq \frac{M}{n} p(x^*) + \frac{1}{n} p^2(x^*)
 \end{aligned}$$

As $M \uparrow$, the variance $\uparrow$ (bias $\downarrow$)

$$\text{MSE} = \text{bias}^2 + \text{variance}$$

$$\leq \frac{L^2}{M^2} + \frac{M p(x^*)}{n} + \frac{1}{n} p(x^*)$$

Take der. of MSE upper bound and set it to 0.

$$\frac{L^2(-2)}{m^3} + \frac{p(x^*)}{n} = 0$$

$$m_{\text{opt}} = \left(\frac{2L^2 n}{p(x^*)} \right)^{1/3} = O(n^{1/3})$$

$$\text{opt. binwidth} = O(n^{-1/3})$$

Optimal MSE for one bin

$$= \frac{L^2}{m_{\text{opt}}^2} + \frac{m_{\text{opt}}}{n} p(x^*) + \frac{p^2(x^*)}{n}$$

$$= L^2 n^{-2/3} + n^{-2/3} p(x^*) + \underbrace{p^2(x^*) n^{-1}}_{\text{circled}}$$

$$= O(n^{-2/3})$$